RAJALAKSHMI ENGINEERING COLLEGE

Department of Computer Science and Design

GE23627 Design Thinking and Innovation

Ideate Phase – Ideation Workshop Sheet

Team Information

Team Name:	Student 1 (Roll No and Name)	Student 2 (Roll No and Name)
MUSIQ (Karaoke n Learn)	241701055 **Sneha S**	241701029 **Mahathi B**

Problem Statement

Singers struggle to enjoy and improve their singing because existing karaoke and music-learning experiences rarely adapt to their individual vocal range, performance needs and learning level, and do not meet the quality expectations of the user.

Karaoke platforms help a user perform a song, while learning tools help a user practise. Very few experiences connect the two around the individual singer — leaving beginners with fixed-pitch tracks they cannot comfortably sing, no way to isolate and repeat difficult phrases, and no objective feedback on how they actually sang.

Target Users

The solution is designed around three tiers of stakeholders identified during the Empathize phase:

Stakeholder Tier	Description and Need
Primary Beginner and amateur singers	Seek confidence and personalized practice to improve their vocal skills; need simple guidance tailored to their own vocal range and learning pace.
Secondary: Music learners and karaoke users	Want easier song practice and flexible performance options so they can enjoy singing while steadily developing their musical ability.
Tertiary : Vocal teachers and music academies	Benefit from structured practice material and objective performance insights that help them support students more effectively between classes.

Brainstorming Rules

1. Focus on generating many ideas.
2. Do not criticize any idea during brainstorming.
3. Encourage creative and unusual ideas.
4. Build on ideas proposed by teammates.
5. Write or sketch ideas.

Idea Generation (Brainstorming)

Idea No	Student 1 Idea (Sneha S)	Student 2 Idea (Mahathi B)
1	Karaoke track generator that separates a song into stems and removes the lead vocal, so any song can become a karaoke track.	Automated voice and performance analysis that scores pitch accuracy, timing, stability and overall performance after each attempt.
2	Vocal range assessment at sign-up that records the user's comfortable low and high notes and builds a vocal profile.	Structured Learn module with alankar practice, breathing exercises and guided song practice with notes and lyrics.
3	Pitch-shifting engine that transposes any song into the singer's comfortable key without distorting audio quality.	Sangathi and phrase breakdown that splits an ornamented passage into learnable units for repeated practice.
4	Section-wise looping with tempo control so a difficult phrase can be slowed down and repeated automatically.	Personalized recommendation engine that suggests exercises, songs and focus areas based on past performance data.
5	Record, playback and export of the user's karaoke performance with independent control over voice and music levels.	Practice history dashboard with difficulty adaptation that raises or lowers challenge as the user improves.

Idea Voting (Dot Voting)

Each member was given ten dots to distribute across the five shortlisted concepts.

Idea	Student 1 Vote	Student 2 Vote	Total Votes
Idea 1 Adaptive Karaoke Track Generator (stem separation + vocal removal)			
Idea 2 Vocal Range Profiling and Personalized Pitch Mapping			
Idea 3 Section-wise Practice Studio (looping, slow-down, sangathi breakdown)			
Idea 4 Voice and Performance Analysis with Scoring			
Idea 5 Personalized Recommendation and Difficulty Adaptation			
Total dots allotted (10 per member)			

Top Idea Selected

Selected Idea: MUSIQ – Karaoke n Learn

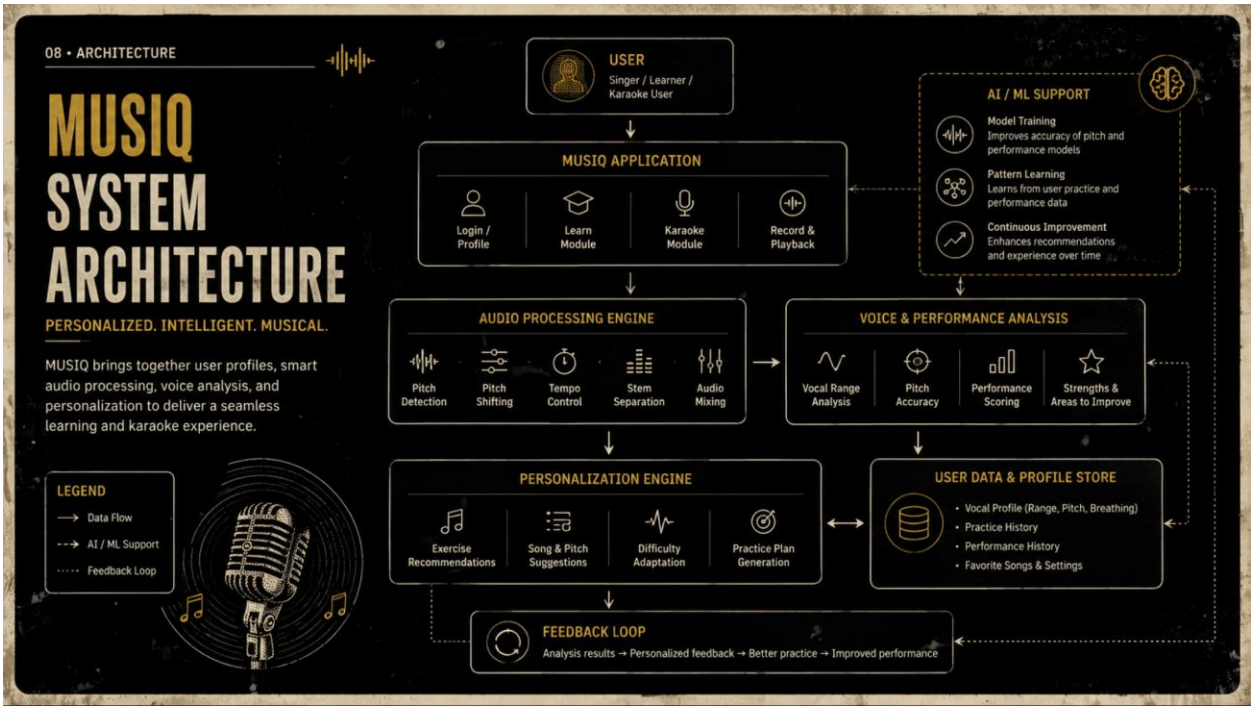
The highest-voted idea, Adaptive Karaoke Track Generator, was taken as the core of the solution, and the next four ideas were merged into it rather than discarded voting showed that the team valued them as parts of one continuous singing journey rather than as separate products.

Concept Development

Feature	Description
Project Title	MUSIQ – Karaoke n Learn
Key Features	• Vocal assessment profiling range, comfortable pitch and breathing at onboarding. • Karaoke generation from any song using stem separation and vocal removal. • Pitch shifting and tempo control to match the user's comfortable key. • Section-wise looping, slow-down and breakdown for hard phrases. • Learn branch: practice, breathing exercises, guided song practice. • Record, playback, save and export with independent voice and music levels. • Performance analysis: pitch accuracy, timing, stability and overall score. • Personalized recommendations and automatic difficulty adaptation.
Technology Required	• Audio source separation to isolate vocals and instruments from a mixed track. • Pitch detection and fundamental-frequency tracking for range and accuracy. • Pitch-shifting and time-stretching libraries for key and tempo changes. • Backend services and a user data store for profiles and practice history.
Expected Benefits	• Any song becomes singable, the track adapts to the singer, not the reverse. • Practice becomes structured and self-directed, without constant supervision. • Objective feedback replaces guesswork, so users know what to improve. • Learning and performing sit in one place, keeping motivation connected. • Confidence grows measurably as difficulty adapts to real improvement.

Solution Architecture Diagram

MUSIQ brings together user profiles, smart audio processing, voice analysis and personalization to deliver a seamless learning and karaoke experience.



Short Idea Pitch

Problem Addressed:

Beginner and amateur singers cannot comfortably sing most songs, because karaoke tracks are fixed in the original singer's key and offer no way to isolate, slow down or repeat a difficult phrase. Learning tools and karaoke platforms exist separately, and neither adapts to the individual's vocal range or gives objective feedback on how they actually sang.

Proposed Solution:

MUSIQ – Karaoke n Learn is a personalized karaoke and vocal-learning platform. It profiles the user's vocal range, converts any song into a customizable karaoke track through stem separation, and shifts pitch and tempo to suit that voice. A Learn branch adds breathing exercises, alankar practice and section-wise song practice, while voice analysis scores every attempt and feeds personalized recommendations back into the next session.

Key Benefit to Users:

Singers get a practice partner that adapts to them instead of the other way around — any song becomes singable in their own key, difficult phrases become learnable, and clear feedback replaces guesswork, so confidence and vocal skill improve measurably over time.

Signature of the Coordinator